SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 7

Financial Analytics

Analytics in finance (sales forecasting, cash planning, credit scoring, fraud, FX risk, investment appraisal). Case 1: monthly sales forecasting with trend, season and Lebaran columns, plus nowcasting. Case 2: credit scoring with logistic regression, tree and random forest — cut-offs and fairness.

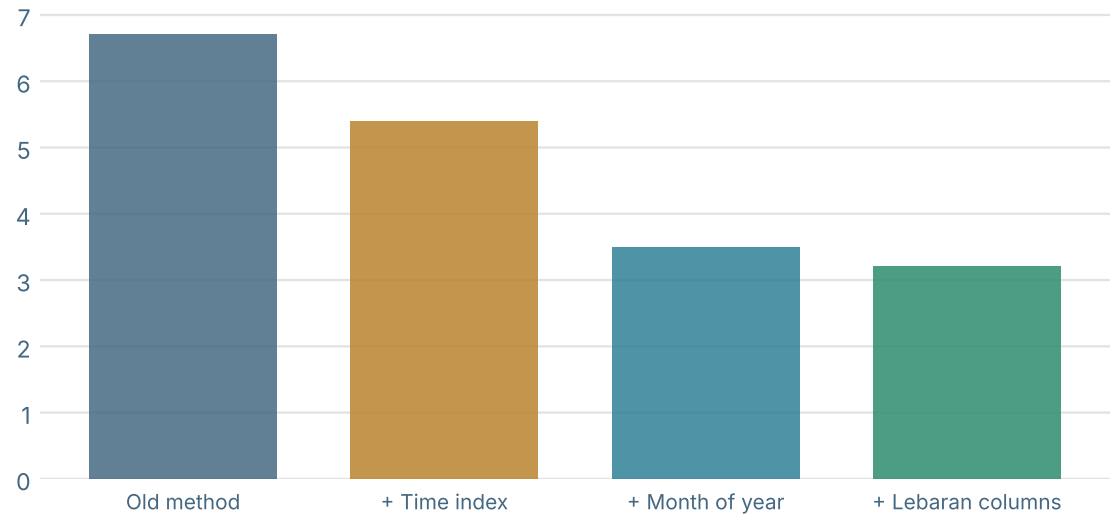
Material

1 Monthly sales forecast for the 2027 budget

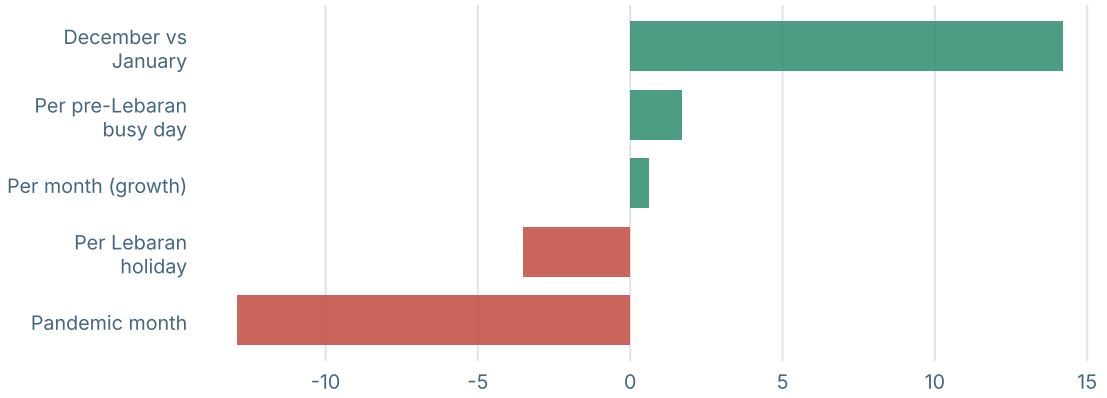
Old method error 6.7%; trend 5.4%; + month 3.5%; + Lebaran busy/holiday days 3.2%. Effects: +1.7% per busy day, −3.5% per holiday day, December +14.2%, +0.6% per month, pandemic −12.9%. For 2027 (Lebaran ~10 March) the peak moves to February — without Lebaran columns IDR 14bn lands in the wrong month.

2 The 2027 budget forecast in charts Deep dive

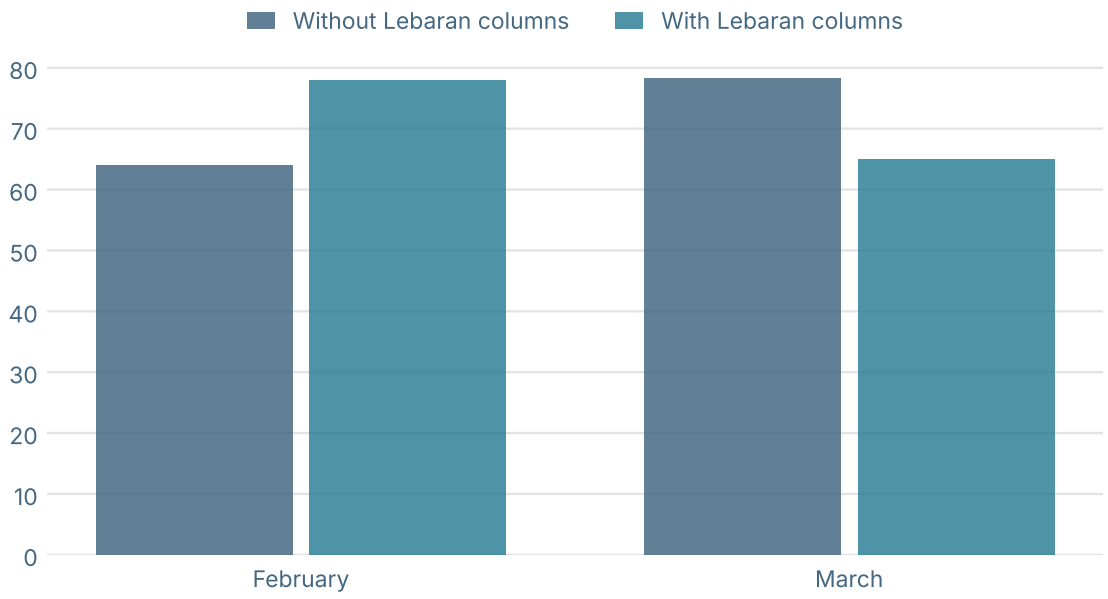
Forecast test error by step (%)



Estimated effect per column (%)



Feb & Mar 2027 forecast (IDR bn)

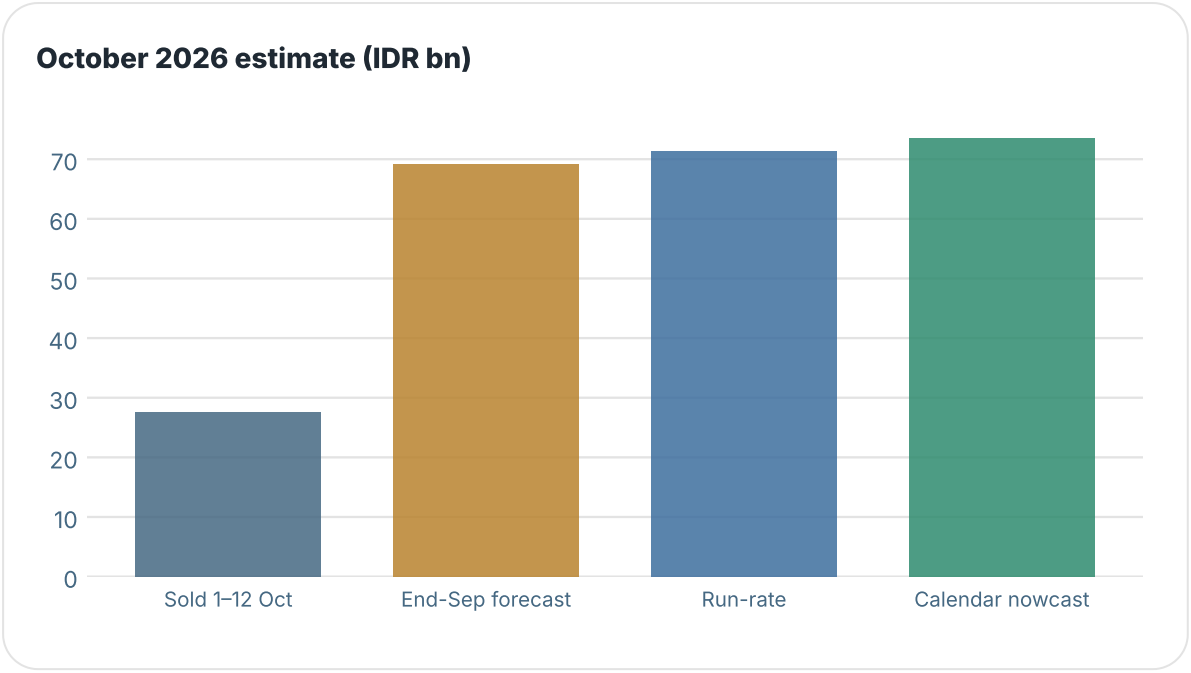


With Lebaran around 10 March 2027, the peak moves to February.

3 Nowcasting the current month

Run-rate 27.6 × 31/12 = 71.3; calendar-based 27.6 / 0.375 = 73.5; forecast 69.2. From day 5 the calendar nowcast beats the forecast.

4 Three ways to estimate October on the 12th Deep dive



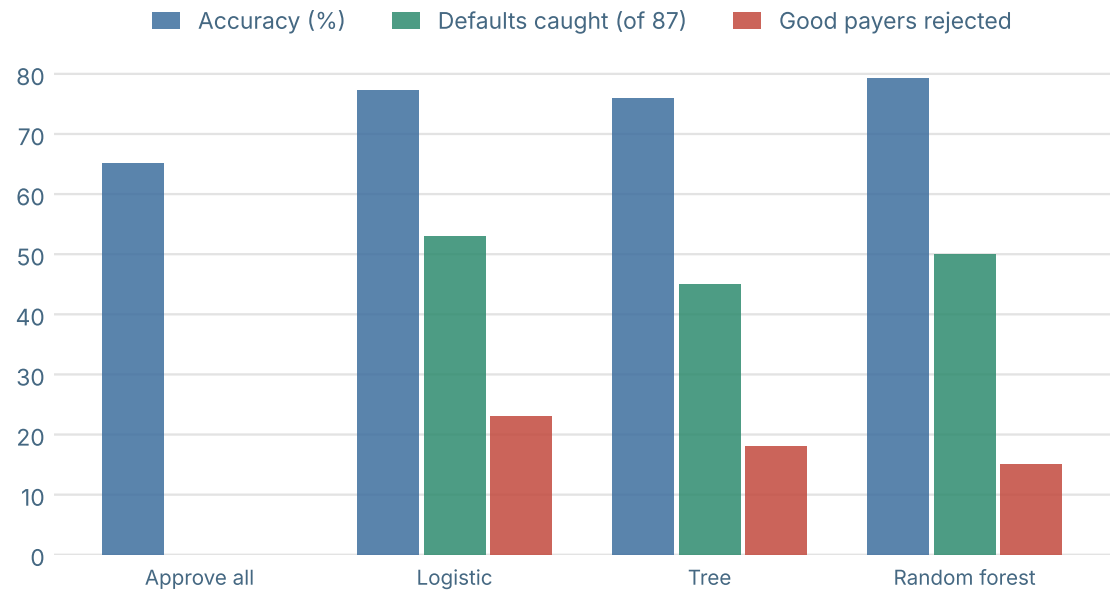
By day 12 a typical October has 37.5% of sales, not 38.7%.

5 Credit scoring

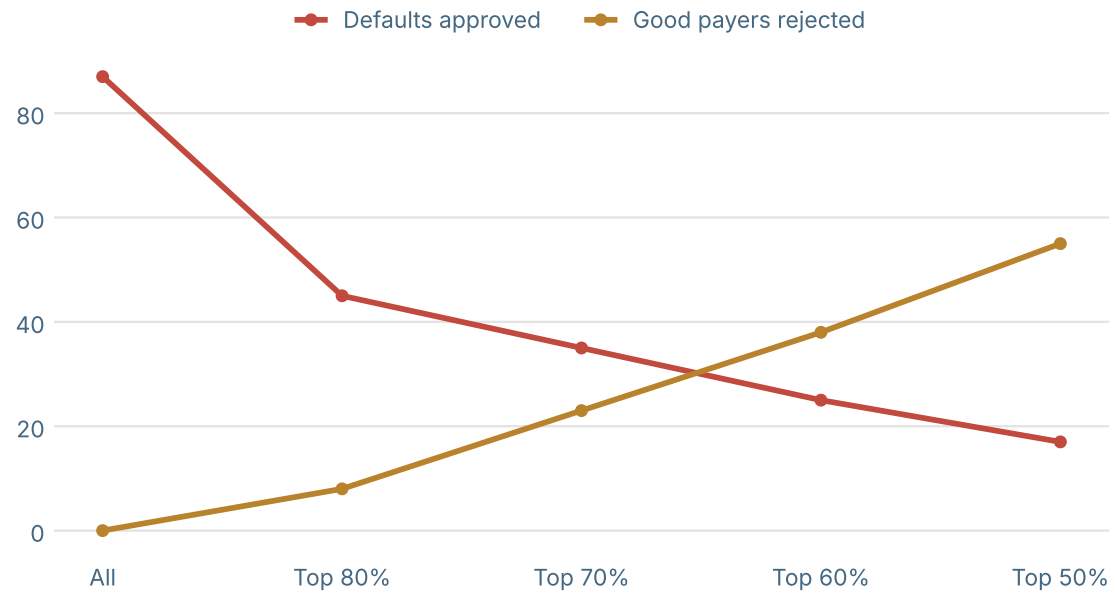
999 applications (34.9% default), 749 train / 250 test. Logistic 77.2% (53/87 defaults caught), tree 76.0%, random forest 79.2% (fewest good customers rejected). Trees read as rules. The model ranks; management sets the cut-off (top 70% approved → 20% default). Using sex adds 3.2 points of accuracy — keeping it is a policy decision.

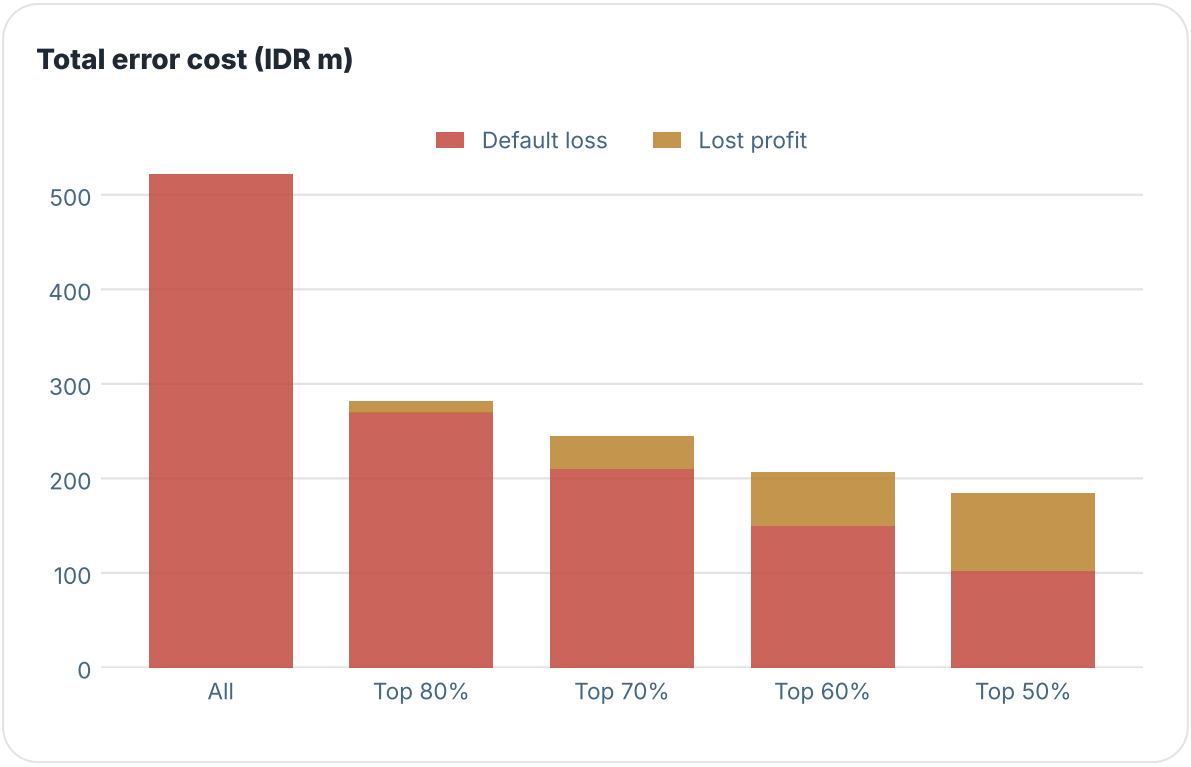
6 Credit scoring: methods and cut-offs in charts Deep dive

Method comparison (250 test cases)



Cut-off trade-off





Top 60% (207m) beats top 70% (244.5m) — and top 50% is cheaper still (184.5m).

Fairness

Women 48.9% vs men 28.7% default here; using gender is an ethical and legal choice.

Key takeaways

1. Finance uses of analytics.
2. Lebaran columns cut error to 3.2%.
3. Moving holidays need their own columns.
4. Calendar nowcasts.
5. Models rank; managers set cut-offs.
6. Sensitive features are policy choices.

Case studies

Case & worked answers

Exercise: scoring a new application and setting the cut-off

A 25-year-old with a 24-month loan falls in an 87.3% default leaf → reject or restructure. With illustrative costs (IDR 6m per default, 1.5m profit per good payer), approving the top 60% costs 207m vs 244.5m for top 70% — cut-offs follow the error costs.

Worked examples

Problem 1 · Easy **Nowcasting**

Sampai tanggal 10, penjualan bulan ini Rp40 M (bulan 30 hari). Data historis: pada tanggal 10 biasanya 30% penjualan bulanan sudah tercapai. Bandingkan run-rate dan nowcast berbasis kalender.

1. Run-rate = 40 × 30 ÷ 10 = 120 (anggap semua hari sama).
2. Kalender = 40 ÷ 0,30 = 133,3.
3. Selisih Rp13 M muncul karena awal bulan biasanya lebih sepi.

Answer: Run-rate Rp120 M; nowcast kalender Rp133 M (lebih akurat bila pola harian stabil).

Problem 2 Medium **Memilih cut-off kredit**

Rugi per default Rp8 juta; laba per pembayar baik Rp2 juta. Top 70%: 30 default disetujui, 20 pembayar baik ditolak. Top 60%: 22 default disetujui, 35 pembayar baik ditolak. Pilih yang mana?

1. Top 70%: 30 × 8 + 20 × 2 = 240 + 40 = 280.
2. Top 60%: 22 × 8 + 35 × 2 = 176 + 70 = 246.

Answer: Top 60% (biaya kesalahan Rp246 juta vs Rp280 juta).

Problem 3 Easy **MAPE**

Aktual 100, 120, 90; forecast 110, 115, 99. Hitung MAPE.

1. |error| ÷ aktual: 10/100 = 10%; 5/120 = 4,17%; 9/90 = 10%.
2. MAPE = (10 + 4,17 + 10) ÷ 3 = 8,06%.

Answer: MAPE ± 8,1%.

Practice questions

Try answering first. The answer key is below each question.

1. Why can't fixed monthly seasonality catch Lebaran?

A. It moves ~11 days a year
B. Always same month
C. No data
D. Pandemic

Answer: A. Lunar calendar.

2. 27.6 sold by day 12; typically 37.5% done. Nowcast…

A. 69.2
B. 71.3
C. 73.5
D. 80.0

Answer: C. 27.6/0.375.

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3. Which method caught the most defaults?

A. Random forest
B. Logistic
C. Tree
D. Approve all

Answer: B. 53 of 87.

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4. Who sets the approval cut-off?

A. The model
B. Management, using error costs
C. Customers
D. Only the regulator

Answer: B. Model ranks; management decides.

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5. Top 60% total cost is…

A. 184.5
B. 207
C. 244.5
D. 282

Answer: B. 150 + 57.

Sources

- Materi P3MD WEEK 7 — Financial Analytics (MK6)